

EUCLEDIAN ALGORITHM

1) $\gcd(25, 60)$

q	r_1	r_2	r
2	60	25	10
2	25	10	5
2	10	5	0
X	5	0	X

($r_1 > r_2$)

$\gcd = r_1 = \underline{\underline{5}}$

$$\begin{array}{r} 2 \leftarrow q \\ 25 \overline{) 60} \\ \underline{-50} \\ 10 \end{array}$$

$$\begin{array}{r} 2 \leftarrow q \\ 10 \overline{) 25} \\ \underline{-20} \\ 5 \end{array}$$

$$\begin{pmatrix} 5 \\ 0 \end{pmatrix} ?$$



(Shortcut method)

M2: $\gcd(25, 60)$

✓

5	25	60
X	5	12

Greatest Common Divisor
 $= 5 = \gcd(25, 60)$

Homework : $\gcd(161, 28)$

[Comment]

✓ M1 / M2

Cryptography
/ Algorithms / I.E.

